ASEN 5090 Lecture 15
Midterm Exam Review

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Course schedule

Week	Date	Topic	Reading	Assignments - Due 11pm*
7	10/5/26	L12. Least Squares Solutions	ME Ch 6.A	
	10/7/26	L13. GPS Point Solutions	ME Ch 6.1	
	10/9/26			
8	10/12/26	L14. Improved GPS Solutions		
	10/14/26	L15. Midterm Review & HW Discussion		
	10/16/26			HW5 Due, HW6 Assigned
9	10/19/26	L16. Precise Positioning Methods	ME Ch 7	
	10/21/26	Midterm Exam (no lecture)		Midterm Release 10/21 8AM
	10/23/26			Midterm Close 10/25 11PM
10	10/26/26	L17. Introduction to Signals & Spectra	ME Ch 8	
	10/28/26	L18. GNSS Spectrum	ME Ch 9	
	10/30/26			HW6 Due, HW7 Assigned

MIDTERM EXAM - Oct 21, 2026 8AM – Oct 25, 2026 11PM

Format – There are three problems worth a total of 100 points.

You may write directly on the exam sheet and/or use your own paper or tablet computer to show your work. Please number all pages and scan them into a single pdf document for upload to Gradescope. All code used must be included in your submitted pdf.

Logistics – The exam period will begin on Wednesday, October 21, 2026 at 8AM and close on Sunday, October 25, 2026 at 11PM. If you have a question about the exam, you may email or text (303-884-1297 include your name) the instructor who will respond promptly to questions between 12-1PM MDT Thursday and Friday and 8-9AM MDT Saturday and Sunday. Once you download the exam from Gradescope, you have up to 24 hours to complete the exam, scan, and upload a pdf of your solutions. However, all solutions must be submitted by the end of the exam period. If you run into difficulty submitting your exam on Gradescope, you may email your solution file to the instructor.

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During the entire exam period (8AM Oct 12 to 11PM on Oct 25) you may NOT work with, consult, collaborate, or discuss the exam with ANYONE except the instructor. Once you have downloaded the exam you may NOT access or search the web, online forums, or resources other than those listed above. Any violation of these exam conditions will result in a grade of F in the course, and an honor code violation report will be filed with the honor code office.

Midterm Exam Topics

- GPS System Basics – requirements, how it works
- Radionavigation Methods – range and doppler measurements
- Signal Structure – carrier, codes (properties), finding a satellite
- Coordinate Frames – ECEF, LLA, ENU
- Orbits – Keplerian elements, GPS constellation, Almanac & Ephemeris
- Clocks & Relativity – Clock stability, how it affects range & doppler
- Visibility – Azimuth & elevation, skyplots, influence on solution accuracy
- Measurements – Pseudorange, carrier phase
- Iono & Tropo – Characteristics, magnitude, how corrected, how observed
- Multipath & receiver noise – Approximate values & how to use combos to see them
- Least squares solutions – General and specific for GPS

Mapping Function: compare tropo & iono

TROPO

$$m(el) = \frac{1}{\sin(el)}$$ Flat surface approximation

$$m(el) = \frac{1}{\sqrt{(1-(cos(el)/1.001)^2}}$$ (ME 5.42)

Separate mapping for dry and wet components

$$m_d(el) = \frac{1}{\sin(el) + \frac{0.00143}{\tan(el)+0.0445}}$$

$$m_w(el) = \frac{1}{\sin(el) + \frac{0.00035}{\tan(el)+0.017}}$$

IONO

$$OF_I = \frac{1}{\cos\zeta'} = \frac{1}{\sqrt{1-\left(\frac{R_E \sin\zeta}{R_E + h_I}\right)^2}}$$

$$OF_I \approx 1.0 + 16.0 \times (0.53 - el)^3$$

Careful with this approximation – el is in semi-circles

GPS LEAST SQUARES POSITIONING

1. Get at least 4 pseudorange (code) measurements from the receiver
2. Guess your location and receiver clock bias

For each observation at a given time (epoch)

3. Compute the expected range for each satellite using information from broadcast ephemeris and your assumed location
4. Compute pseudorange corrections - satellite clock error, relativity, iono, tropo
5. Compute pre-fit measurement residual (scalar)
6. Compute measurement sensitivity matrix (1x4)

Use all the observations to solve for position correction

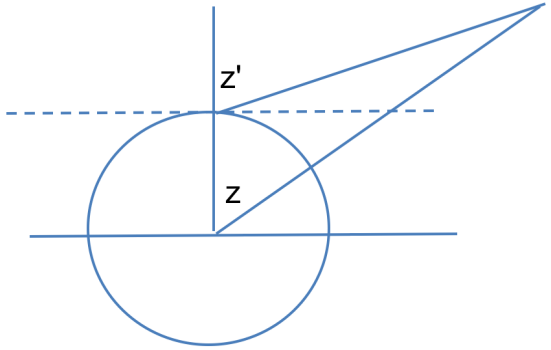
7. Assemble measurement residuals sensitivities into arrays
8. Solve for a correction to assumed location and clock bias

If necessary, repeat steps 3-8!

The coordinate frame you choose in defining the G matrix, determines the coordinate frame of your solution.

Viewing Geometry Notes

- Visibility Limits (spherical Earth)



At the North Pole:

$$\tan z' = \frac{R_{GPS}\sin z}{R_{GPS}\cos z - R_E} = \frac{26{,}610 * \sin(35)}{26{,}610\cos(35) - 6378}$$

$z' = 44.7\,\text{deg}$ smallest zenith angle at NP

el' = 90-z' = 45.29 deg highest elevation at NP

At the Equator due North

$$\tan z' = \frac{R_{GPS}\sin inc}{R_{GPS}\cos inc - R_E} = \frac{26{,}610 * \sin(55)}{26{,}610\cos(55) - 6378}$$

$z' = 67.82\,\text{deg}$ maximum zenith angle due north

el' = 90-67.82 = 22.17 min el angle due north

Range as a function of elevation angle

- For spherical Earth (R_E), circular GPS orbit (R_{GPS}), and satellite elevation (*el*)

- Derive this:

$$R = \sqrt{R_E^2(\sin^2 el - 1) + R_{GPS}^2} - R_E \sin el$$

ME Eq 10.2

Check to see that it makes sense for elevations of 90 deg and 0 deg.

Range-rate/Doppler as a function of elevation angle?

- More complicated because velocities of GPS and observer are not coplanar with the position vectors.
- What is the smallest (magnitude) range-rate you should expect to see on a GPS pass? What is the largest you could see?

Acknowledgements

ASEN5090 lecture materials have been developed at the University of Colorado Boulder by Professors Penina Axelrad, Dennis Akos, Kristine M. Larson, Jade Morton, and Yang Wang, with additional contributions by many outstanding TA's and students.

Next lecture: Precise Positioning Methods

Enter the start and end date/time of your exam:
Sign the honor code pledge below before submitting your exam.

Start Date & Time: _____

End Date & Time: _____

ASEN 5090 Introduction to GNSS

MIDTERM EXAM

October 16, 2025 8AM – October 19, 2025 11PM (MDT)

Format – There are three problems worth a total of 100 points.

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The name of your pdf file should be: *LastName_FirstName_ASEN5090Midterm.pdf*

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If doing calculations by hand, please assume that the speed of light, $c = 3 \times 10^8$ m/s, $\mu_E = 398600.5$ km^3/s^2. If you are using existing code, you may use these values or other established values - make sure to clearly identify the values used in your calculations.

On my honor, as a University of Colorado Boulder student, I have followed all of the above instructions and have neither given nor received unauthorized assistance on this work.

Signature Date

Printed name

PROBLEM 1. (40 points, a-d)

For this problem, assume GPS orbits are circular with radius R_{GPS} = 26,560,000 m and inclination= 55 deg. The table below gives the WGS-84 ECEF locations of a few satellites in circular orbits (not necessarily GPS satellites) at a particular time.

a) Compute the magnitude of the velocity for each of the satellites in the table below, assuming they are all in circular orbits. Which satellite would have the largest possible range rate (and largest magnitude Doppler shift) when tracked by observers on the Earth?

b) An observer (A) is located at N 26 deg, longitude W 80 deg, height above the ellipsoid h=0. Compute the WGS-84 ECEF coordinates of A in meters. Express the results (x_A, y_A, z_A) to 1m precision.

c) Compute the geometrical range, azimuth, and elevation for each satellite position in the table from observer A's location. Add the appropriate units for these quantities to the table. (*It is not necessary to account for signal time-of-flight in this problem*.)

d) Which satellites would you expect to be visible to observer A (mark them with a V)?
Which satellite is closest to the observer's zenith (mark it with a Z)?
Which satellites which could be GPS satellites (mark them with a G)?
Show your work & briefly explain your results – *required to get full credit for correct answers*.

	ECEF X (m)	**ECEF Y (m)**	**ECEF Z (m)**	**|v|**	**Range to A**	**Azimuth**	**Elevation**
Sat 1	1,193,698	-4,454,942	26,156,494				
Sat 2	832,273	-4,720,052	4,792,867				
Sat 3	-3,994,192	22,652,188	13,280,000				
Sat 4	21,189,069	-36,700,543	0				
Sat 5	7,867,022	-21,614,466	13,280,000				

PROBLEM 2. (30 points, parts a-d)

The figures on the following page show the 24-hr skyplot and a 4-hr elevation vs time plot for a location somewhere on the Earth. The elevation vs time plot indicates the specific PRN at its highest elevation. Assume the GPS orbits are circular with radius R_{GPS} = 26,560 km and the Earth radius R_E = 6378 km.

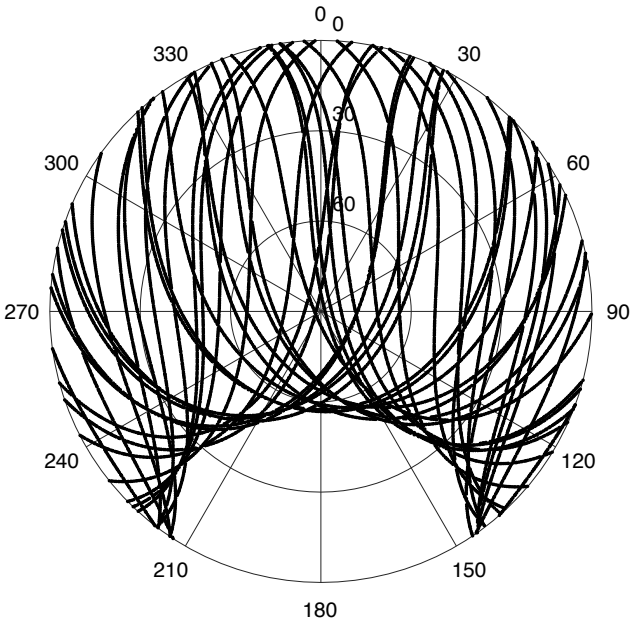
a) What do you know about the latitude of the location? Explain

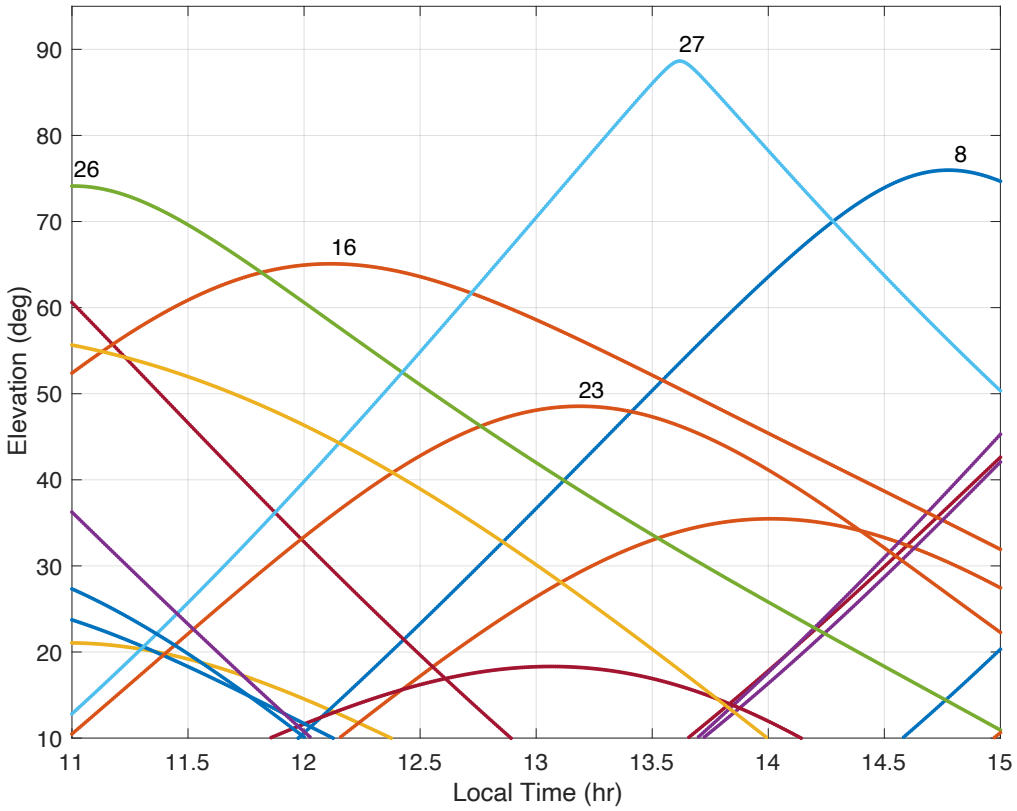
b) Sketch the approximate range from the observer to PRN 27 over this 4-hour period and give approximate max and min values. (Label your axes.)

c) At what time within this 4 hour period does PRN 16 have a zero Doppler shift? At what time in this interval does it have its largest positive Doppler shift? Explain how you identified these times.

d) Which satellite would you expect to have the smallest ionospheric delay at t=13:00 hours? Which one is likely to have the largest ionospheric delay? Why?

24 Hour Visibility

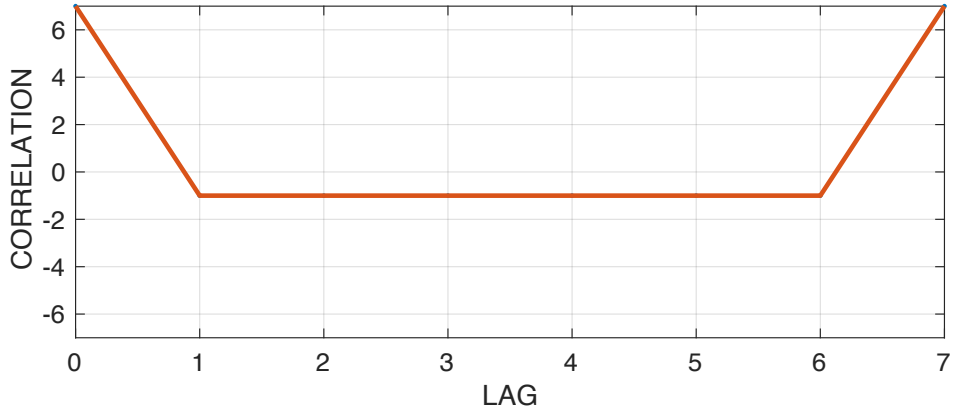




PROBLEM 3. (30 points, a-d)
Two PRN codes, A and B, are considered for a mini ranging system.

```
A = [ 1   1   1 -1 -1   1 -1 ]
B = [ 1 -1 -1   1   1 -1   1 ]
```

The Autocorrelation for Code A is shown below:



a) Compute the auto-correlation for Code B and plot it on the graph above.

b) What is special about Code A compared to Code B? Is Code A useful for ranging? Explain your answer.

c) Would this pair of codes be useful for CDMA? Why or why not?

d) If the code chipping rate is 2 MHz:

- What distance (in meters) corresponds to a shift of 1 chip of the A-code?

- How long (in seconds) does it take to transmit the entire A-code?

Enter the start and end date/time of your exam:
Sign the honor code pledge below before submitting your exam.

Start Date & Time: _____

End Date & Time: _____

ASEN 5090 Introduction to GNSS

FINAL EXAM

December 5, 2025 6PM – December 10, 2025 10PM (MST)

Format – There are two problems (each with multiple parts) worth a total of 100 points.

Submissions will include tables to fill out on the exam sheet and additional pages (handwritten or typed) answering questions, showing your work and all code used for calculations. Submission to Gradescope will be a single pdf document with pages assigned to the correct problem. Answers without supporting work will not receive credit.

Logistics – The exam period will begin on Friday, Dec 5, 2025 at 6PM and close on Wednesday, Dec 10, 2025 at 10PM (MST). If you have a question about the exam you may email or text (303-884-1297 include your name) the instructor who will respond promptly to questions on Monday-Wednesday between 10-11am.
Once you download the exam from Gradescope, you have up to 24 hours to complete the exam, scan, and upload a pdf of your solutions to Gradescope. However, all solutions must be submitted by the end of the exam period. If you run into difficulty submitting your exam on Gradescope, you may email your solution file to the instructor – but you will still be subject to the 24-hr and end of exam period limits.

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During the entire exam period (6PM Dec 5 to 10PM Dec 10) you may NOT work with, consult, collaborate, or discuss the exam with ANYONE except the instructor. Once you have downloaded the exam you may NOT access or search the web, online forums, or resources other than those listed above. Any violation of these exam conditions will result in a grade of F in the course and an honor code violation report will be filed with the honor code office.

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Signature Date

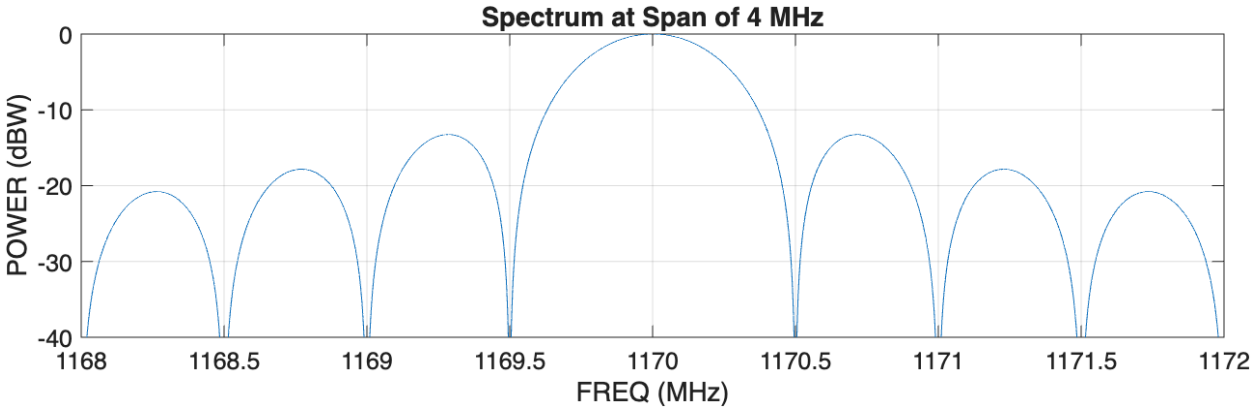
Printed name

PROBLEM 1. (50 points, parts a & b)

Figure 1 illustrates the spectrum for a proposed new navigation system. The signals are modulated with a maximal length PRN code generated with a 12 stage shift register. The satellites for this new system are in low Earth orbit at an orbital altitude of 850 km. Assume a spherical Earth with radius 6378 km.

a) Determine the carrier frequency, carrier wavelength, code chipping rate, and length of the entire PRN code. **Fill in the values in the table below** and show/explain your calculations in the space provided or on additional sheets.
In Figure 1, add a sketch of the signal spectrum that would be observed if the new system combines the code shown with a square wave to produce a BOC(1,1) signal.

	Value	Units
Carrier frequency		
Carrier wavelength		
Code chipping rate		
Full code length or duration		



b) Consider two receivers on the Earth surface, “Z” is directly below the transmitting satellite and “H” sees the satellite just rising along the horizon. Use the information given in the problem statement and figure in part (a), and the values given in the table below to compute results for each of the remaining table entries. **Fill in the values in the table below** and show/explain your calculations in the space provided or on additional sheets. There is no need to consider time-of-flight adjustments to the range or any other complicating factors for these calculations.

	Observer Z (zenith)	Observer H (horizon)	Units
Satellite elevation			deg
Range			km
Doppler Shift			Hz
Space Loss			dB
C/No (*given*)	50 dB-Hz	37 dB-Hz	dB-Hz
Code Tracking sigma (*given for Z*)	0.1 m		m

PROBLEM 2. (50 points, parts a-f)

Five data files are provided for September 14, 2025

1) composite broadcast ephemeris file for this date
2 & 3) .mat files with GPS observations for a single epoch (time) for a MYSTERY location and for NIST
3 & 4) rinex ascii versions of the observation files for people whose code is set up to read this in directly

Use the data provided to solve for the mystery location. Enter your ECEF position and clock bias results for each iteration (up to 5) in the table below with precision of 0.1 m. **Attach your code and answer the questions below.**

If you are not able to successfully solve for MYST, you may submit results for NIST for partial credit.

Iteration	X (m)	Y (m)	Z (m)	Clock bias (m)
Initial Guess				
1				
2				
3				
4				
5				
Final answer in LLH	Lat:	Lon:	Ht:	

a) Please use the C1C and C2W pseudoranges to compute your solution.
Why do we use two different pseudoranges for positioning?

b) Choose an initial guess for the location and justify this choice.

c) Choose an elevation cutoff and justify this choice.

d) Choose a tropospheric model to use and a reasonable zenith delay value.

e) How did you determine if your position & clock solution is correct?
Describe the convergence of your iteration.
Give final values for the post-fit residuals for each satellite.

f) What do you notice about the MYST observations compared to what you see with the NIST data in HW5 (or with the attached file)? Compute the HDOP and VDOP for the MYST solution and compare these values to the DOPs for a NIST solution. (You can compute a solution using the NIST file provided, or refer back to your NIST results for HW5.) Briefly explain why they are similar or significantly different.

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Start Date & Time: _____

End Date & Time: _____

ASEN 5090 Introduction to GNSS

FINAL EXAM

Due December 11, 2022 4 pm, in Gradescope

Format – There are two problems worth a total of 100 points.

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The name of your pdf file should be: *LastName_FirstName_ASEN5090Final.pdf*

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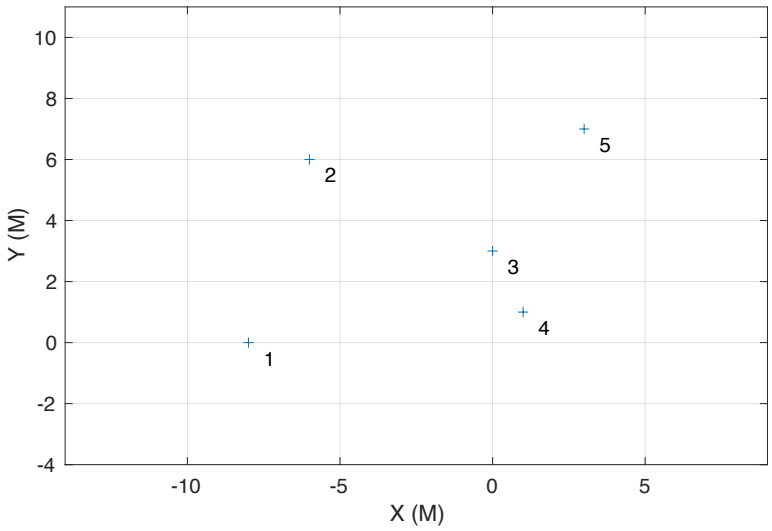
Signature Date

Printed name

PROBLEM 1. (50 points, a-f)

Consider a miniature two dimensional GPS-like system in which the receiver and all transmitters are located in the plane of the paper within the area shown in the figure. The table and figure below give the locations of 5 transmitters and the corresponding pseudorange measurements made by a receiver at an unknown location also within the area shown in the figure. Assume that all measurements have the same quality and that the only systematic error source is the receiver clock.

PRN	Satellite Position (m)	Pseudorange (m)
1	X1 = -8.0 Y1 = 0.0	17.0
2	X2 = -6.0 Y2 = 6.0	15.5
3	X3 = 0.0 Y3 = 3.0	15.9
4	X4 = 1. 0 Y4 = 1.0	17.4
5	X5 = 3.0 Y5 = 7.0	20.2



a. Select a reasonable initial guess for the receiver location and clock bias, and explain your reasoning.

b. Using the given information and your initial guess, set up a least squares solution to estimate the receiver location and briefly describe the procedure for solving it. Give the initial measurement sensitivity matrix (A matrix) and prefit residuals.

c. Solve for the receiver location and clock bias. Is it necessary to iterate? Does your solution converge? If so, how may iterations are required?

d. How important is it to have an accurate initial guess in this situation compared to the regular GPS geometry? Why?

e. Compute the HDOP using all 5 transmitters (based on your final result).

f. Choose what you think would be a particularly good or bad subset of three transmitters and recompute a new solution and HDOP. Explain the reasoning for your choice and whether your results make sense compared to the 5 measurement case above.